

Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

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Abstract—We present a blind disturbance-model selector for diffusion-aided semantic image reception. Given a received latent, the selector uses excess energy and block-energy heterogeneity to choose Gaussian denoising or joint signal–interference reconstruction before diffusion inference. This choice addresses a practical limitation of joint reconstruction: estimating an unknown interferer can add both reconstruction error and computation. We fit the rule on separate development images and then freeze it together with all neural weights. Only the selected receiver is executed. We evaluate the rule on CelebA images with CIFAR-10 interference, using identical observations for all receiver comparisons and a same-gate joint receiver as a control. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule reaches 21.593 dB mean PSNR. It improves on calibrated joint sampling by +1.334 dB and on blind Gaussian denoising by +0.750 dB. The rule uses 52.2 predictor evaluations per frame on average, compared with 80 for joint sampling. Separately trained DeepJSCC and MambaJSCC codecs provide external performance references under the same channel-use budget.

Index Terms—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source–channel coding.

I. INTRODUCTION

We consider semantic image reception under co-channel interference of unknown power and structure. Deep joint source–channel coding (JSCC) maps images directly to channel symbols and degrades gradually as channel noise increases [1]. SwinJSCC and MambaJSCC implement this principle with transformer and state-space architectures [2], [3]. Even so, a deployed codec must recover images from unmodeled interference without changing the transmitted latent or adding unnecessary receiver computation.

Generative receivers use learned priors to recover corrupted latents. CDDM matches the aggregate disturbance to a diffusion noise level and denoises the received latent [4]. ICDM instead reconstructs the desired and interfering latents jointly [5]. Iterative source error correction [6], channel-conditioned diffusion reconstruction [7], and blind inverse-problem inference [8] provide related forms of receiver adaptation. These methods provide recovery mechanisms, but they do not resolve which disturbance model a receiver should use for a particular observation.

Joint reconstruction is preferable only when the recovered interference structure offsets its estimation error. The receiver must otherwise infer two latents and an unknown amplitude from one mixture using imperfect priors and finite-step sampling. An inaccurate interference estimate can distort the de-

sired latent and still consume twice the predictor evaluations. Gaussian denoising avoids this burden, but discards structure that can aid recovery. We therefore ask whether the receiver can exploit joint reconstruction only when the received frame supports its additional model assumptions.

Our key observation is that a richer disturbance model contributes both structure and estimation uncertainty. The preferred model can therefore vary from frame to frame. Excess received energy measures aggregate disturbance relative to the known signal and noise powers. Block-energy variation measures how unevenly that energy is distributed. Both statistics are available before diffusion inference and do not require identifying the interfering image. Because desired-signal variation also affects them, their value must be established on held-out images.

We address this decision with a blind selector based on the two statistics. Image-group validation compares energy-only and two-feature threshold rules on separate development images. It retains a fixed receiver when selection gives no validation gain. The resulting rule is frozen and executes one branch per frame. All neural weights remain fixed, isolating receiver selection from codec or prior retraining. The experiment therefore asks a narrow question: whether observation-dependent receiver choice improves a fixed generative receiver stack.

On 256 independent image pairs, the selector improves mean PSNR by 0.750 dB over blind Gaussian denoising and 1.334 dB over calibrated joint sampling. It also uses 34.7% fewer predictor evaluations than joint sampling. A same-gate control distinguishes model selection from merely bypassing joint inference, while harmful decisions and block mismatch expose its limits. Separately trained DeepJSCC and MambaJSCC codecs provide external reference points. The shared-prior receiver ports do not reproduce the complete published CDDM or ICDM systems.

To summarize, we make three contributions. (1) We formulate disturbance-model selection as a blind reconstruction decision for a frozen semantic codec. (2) We develop an $O(n)$ energy-based rule that selects one receiver before generative inference without neural retraining. (3) We evaluate the frozen rule on paired independent observations, establishing its mean-quality and computation benefit while measuring harmful decisions and mismatch sensitivity. Sections II and III present the system model and selector; Sections IV and V report experiments and conclusions.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we first model image transmission, then introduce the two diffusion-assisted recovery models, and finally formulate receiver selection.

A. Image Representation and Transmission

Let $\mathbf{u} \in [0, 1]^d$ be an RGB image, with $d = 3H_{\text{img}}W_{\text{img}}$. The JSCC encoder f_s combines representation learning and channel mapping; f_z encodes an interfering image $\mathbf{v}$ into the same dimension:

$$\mathbf{x} = \mathcal{N}(f_s(\mathbf{u})), \quad \mathbf{z} = \mathcal{N}(f_z(\mathbf{v})), \quad \mathcal{N}(\mathbf{t}) = \frac{\sqrt{n} \mathbf{t}}{\|\mathbf{t}\|}. \quad (1)$$

Each nonzero codeword contains n complex symbols with unit average power, giving bandwidth ratio n/d . We partition the symbols into B blocks of lengths L_b , where $\sum_b L_b = n$. The received symbols are

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0. \quad (2)$$

Here a_b denotes the unknown interference amplitude in block b , and N_0 denotes the complex noise power. We assume unit desired-channel gain and known symbol alignment. The receiver knows N_0 , the normalization, and the block boundaries, but not $\mathbf{u}$, $\mathbf{z}$, or a_b . For nominal SINR γ , the simulator sets the average interference power to $10^{-\gamma/10} - N_0$. Realized block energies of $\mathbf{z}$ make the per-frame SINR differ from this nominal setting.

B. Diffusion-Assisted Image Recovery

The receiver combines the frame-specific evidence in $\mathbf{y}$ with learned latent priors. Joint reconstruction uses both priors:

$$p(\mathbf{x}, \mathbf{z} \mid \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x})p_z(\mathbf{z}) \times \exp \left[-\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2 \right]. \quad (3)$$

The amplitudes must be estimated. Moreover, a matching mixture cannot identify both latents: $(\mathbf{x}_b + a_b \mathbf{z}_b, \mathbf{z}_b - \mathbf{z}_b)$ produces the same observation. Gaussian recovery instead uses only p_s :

$$p_G(\mathbf{y} \mid \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (4)$$

The Gaussian and joint receivers, $\mathcal{R}_G$ and $\mathcal{R}_S$, use frozen diffusion priors with observation-based initialization or guidance. Neither performs exact posterior sampling, and only joint recovery estimates a second latent. A blind rule $\pi_\theta(\mathbf{y}) \in \{G, S\}$ selects one branch. The common decoder then reconstructs

$$\hat{\mathbf{u}}_\theta = g_s(\mathcal{N}(\mathcal{R}_{\pi_\theta(\mathbf{y})}(\mathbf{y}))). \quad (5)$$

C. Quality, Cost, and Receiver Selection

We measure image reconstruction with per-image PSNR, $q(\mathbf{u}, \hat{\mathbf{u}}) = -10 \log_{10}(\|\mathbf{u} - \hat{\mathbf{u}}\|^2/d)$ for clipped RGB outputs. LPIPS and MS-SSIM provide additional fidelity measures; they do not measure downstream task accuracy. We measure computation by counting calls to both priors, and receiver latency includes selection and decoding.

For development pairs $\mathcal{D} = \{(\mathbf{u}_i, \mathbf{y}_i)\}_{i=1}^M$, we fit

$$\theta^* \in \arg \max_{\theta \in \Theta} \frac{1}{M} \sum_{i=1}^M q(\mathbf{u}_i, \hat{\mathbf{u}}_\theta(\mathbf{y}_i)), \quad (6)$$

where Θ contains the rule family retained by image-group validation and a constant-receiver fallback. Bandwidth, power, and neural weights remain fixed; we report computation separately from the PSNR objective. This fitting is offline. At inference, the frozen rule executes one branch without access to its reconstruction error. Section III develops the observable decision.

III. BLIND DISTURBANCE-MODEL SELECTION

In this section, we first describe the two candidate receivers and then develop a blind rule that selects one before diffusion inference.

A. Candidate Receivers

Both receivers share the frozen desired-signal prior and decoder. The Gaussian receiver first estimates aggregate disturbance from the received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (7)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and a vanishing signal–interference cross expectation, unclipped excess energy estimates the mean interference energy. Finite blocks, latent correlation, and clipping make this estimate inexact. In particular, local desired-signal energy can resemble interference heterogeneity.

Let $r(\mathbf{y}) \in \mathbb{R}^{2n}$ stack real and imaginary components. Given the training diffusion schedule $\bar{\alpha}_t$, the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (8)$$

The factor 1/2 converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor ϵ_s gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (9)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (10)$$

The receiver uses at most 40 predictor evaluations and fewer when the matched initial index is below 39.

The joint receiver S initializes $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$ and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$, then updates

$$a_b^+ = \left[\frac{\Re\{\tilde{\mathbf{z}}_b^H(\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (11)$$

An empty denominator retains the previous amplitude. This nonnegative least-squares update is conditional on point estimates and does not correct their uncertainty. Calibration

begins when the signal coefficient $\alpha(t) \geq 0.5$. Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. The existing 40-step, order-three sampler calls each prior 40 times. An initialization-only variant keeps the amplitudes fixed and isolates the effect of calibration.

B. Selection Before Generative Inference

The selector combines $\hat{P}$ with normalized block-energy variation:

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \varepsilon)}. \quad (12)$$

Here $\varepsilon > 0$ prevents division by zero, and all receiver blocks have equal length. The statistic H measures energy variation, not the interference distribution. For $\theta = (\tau_\ell, \tau_h, \eta)$, the rule is

$$\pi_\theta(\mathbf{y}) = \begin{cases} \text{G}, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ \text{G}, & \hat{P} \leq \tau_h, H > \eta, \\ \text{S}, & \text{otherwise.} \end{cases} \quad (13)$$

We also test the reversed assignment of G and S in (13); thus, the Gaussian branch is not assigned to the low-power region a priori. The energy-only family sets $\tau_\ell = \tau_h$. Either family can reduce to a constant receiver, allowing validation to reject adaptation.

We fit the thresholds to mean per-image PSNR on development data. Eight folds keep all channel conditions for an image together. For each fold, thresholds are fitted on the other seven folds and evaluated on the held-out images. We then refit the family with the highest out-of-fold mean on all development images. Ties favor lower inference cost and then the simpler family. If neither family improves on the better fixed receiver, validation retains that receiver. Confirmation and pressure results never alter the thresholds. At inference, the selector evaluates (13) and executes only the chosen branch. This adds $O(n)$ energy calculations and no diffusion evaluations.

For analysis, let $q_j(\mathbf{u}, \mathbf{y})$ be a candidate's PSNR. The unavailable per-frame oracle has quality $\max(q_G, q_S)$. The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (14)$$

shows that complementary receivers alone are insufficient: errors on large-gap frames dominate regret. This identity is an accounting relation, not a generalization bound. The fitting objective is mean PSNR; it does not optimize mean MSE, LPIPS, or harmful-tail probability.

IV. EXPERIMENTAL EVALUATION

In this section, we compare reconstruction quality and inference cost with fixed receivers and external codecs. We then examine harmful decisions and interference mismatch.

A. Experimental Setup

We use 128×128 CelebA center crops as desired images [9]; 32×32 CIFAR-10 interference images are nearest-neighbor upsampled. Training uses 162,770 desired and 50,000 interference images. The frozen SwinJSCC codec has stage widths (128, 192, 256), depths (2, 2, 6), and an $8 \times 16 \times 16$ real latent: 1,024 complex symbols at CBR 1/48. Each DiT prior has 28 blocks, width 1,152, 16 heads and patch size two. The inherited training schedule uses 40/100 desired/interference-codec epochs and 50/20 desired/interference-prior epochs; selection adds no neural training.

For external references, DeepJSCC and MambaJSCC each train desired and interference codecs for 40 epochs at SNR 20 dB and CBR 1/48, using one seed and validation MSE for checkpoint selection. DeepJSCC uses a pinned third-party PyTorch reproduction; MambaJSCC uses official code. Public checkpoints are also evaluated without fine-tuning, with one encoder processing both image streams. Public MambaJSCC differs in training domain, channel and resolution; public DeepJSCC additionally uses CBR 0.16256. We treat these results as transfer diagnostics. The SwinJSCC-specific diffusion priors cannot be attached to either codec without new prior training. We do not reproduce the complete published CDDM or ICDM systems.

Development uses 64 validation pairs and seed 1700. For independent confirmation, we use 256 test pairs, seeds 2700/2701, SNR 20 dB, and nominal SINRs $\{-7, -4, 0, 4, 7, 10\}$ dB. Sixteen 64-symbol blocks have stationary, alternating (0.25, 1.75), or burst (0, 2) relative interference powers. These 18 conditions give 9,216 frames per receiver. We run confirmation only after freezing the policy. File hashes exclude overlap with development, pressure tests, and 200 previously evaluated pairs; independence is at image-pair, not identity, level. The power grid is $\{0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8, 10^6\}$; the heterogeneity grid is $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$, giving 26 energy-only and 2,028 two-feature rules across both assignment directions.

We compare with direct decoding, Gaussian reception using N_0 alone, blind Gaussian reception, initialization-only and calibrated joint sampling, and the selector. The same-gate control uses direct decoding for $\hat{P} \leq 0.15$ and calibrated joint sampling otherwise. Its outputs and component times are reconstructed from paired branches, excluding gate overhead. Development also tests coupled feedback and a Gaussian receiver given nominal disturbance power, a privileged diagnostic. All SwinJSCC variants share received tensors and sampler seeds; external codecs share image pairs, conditions and channel seeds but generate their own codewords.

We measure all receivers in float32 with batch size one, TF32 disabled, four CPU threads, and an RTX 4090. Synchronized receiver time includes selection, sampling, and decoding; encoding, metric computation and loading are excluded. The number of function evaluations (NFE) counts calls to both priors. Quality metrics are per-image PSNR and MSE, LPIPS-

TABLE I
RECONSTRUCTION QUALITY AND RECEIVER COST ON THE
CONFIRMATION SET.

Receiver	PSNR	MSE $\times 10^3$	LPIPS	MS-SSIM	NFE	ms
SwinJSCC direct	18.67	29.14	0.440	0.697	0.0	8
DeepJSCC direct	17.39	24.71	0.575	0.684	0.0	1
MambaJSCC direct	16.92	44.13	0.463	0.613	0.0	10
Gaussian (N0)	18.83	28.70	0.438	0.700	18.0	225
Blind Gaussian	20.84	24.90	0.378	0.782	40.0	488
Init. joint	19.61	20.15	0.363	0.793	80.0	1110
Calibrated joint	20.26	18.49	0.362	0.802	80.0	1180
Joint + gate	20.45	18.43	0.361	0.801	65.9	974
Selector	21.59	16.45	0.336	0.832	52.2	701

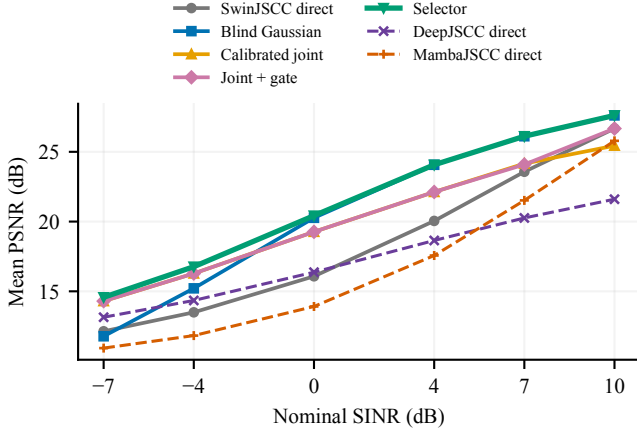


Fig. 1. Mean PSNR versus nominal SINR over the three interference profiles.

VGG v0.1 [10], and four-scale MS-SSIM [11], averaged arithmetically after RGB clipping to $[0, 1]$; four scales fit the image resolution. Paired 95% intervals use 5,000 bootstrap resamples of image pairs, retaining both seeds and all conditions. A harmful decision loses more than 0.5 dB against its comparator.

B. Frozen Rule and Receiver Comparison

Image-group validation compares the two rule families. Energy-only selection yields 22.002 dB, whereas the two-feature family yields 22.239 dB. We therefore retain the energy and heterogeneity rule with $\tau_\ell = 8$, $\tau_h = 0.5$ and $\eta = 0.2$, with Gaussian reception below the applicable power threshold.

The low-heterogeneity development region contains 810 frames and has a maximum excess-energy estimate of 5.241. The fitted threshold 8 therefore lies beyond observed support and makes this region always Gaussian. This threshold is not an identified physical transition, and the family comparison does not establish that either feature is individually necessary. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development comparisons motivate the receiver decision but are not independent confirmation. Table I and Fig. 1 summarize independent confirmation. The selector changes mean PSNR by +1.334

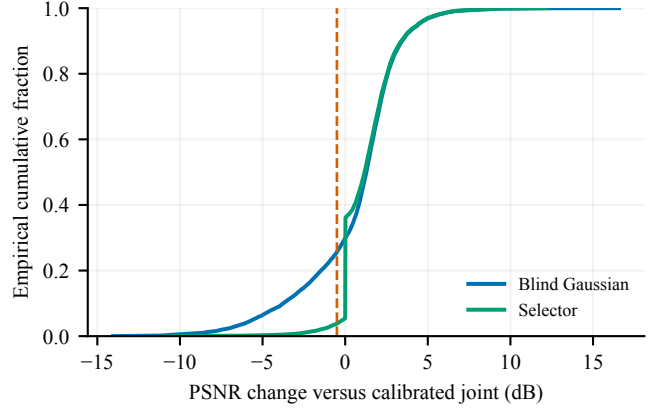


Fig. 2. Empirical CDF of per-frame PSNR change relative to calibrated joint sampling.

dB relative to calibrated joint sampling (95% interval $[1.265, 1.404]$), +0.750 dB relative to blind Gaussian denoising (95% interval $[0.698, 0.806]$), and +1.140 dB relative to the same-gate control (95% interval $[1.075, 1.205]$). Both fixed-receiver comparisons have positive paired intervals. This supports a mean-PSNR benefit within the matched confirmation population. Estimating total disturbance changes the Gaussian receiver from 18.829 to 20.843 dB, so the thermal-noise-only port is insufficient as the sole Gaussian baseline.

The fixed receivers are complementary across interference profiles. Under stationary interference, Gaussian denoising reaches 21.310 dB, compared with 19.282 dB for joint sampling. Under burst interference, joint sampling reaches 21.317 dB, compared with 20.481 dB for Gaussian denoising. The selector reaches 21.317, 21.285, and 22.177 dB for stationary, alternating, and burst interference, respectively. These results are consistent with using different receivers for different observations. They do not establish a physical interpretation of the energy statistic.

Separately matched direct-codec anchors reach 17.394 dB for DeepJSCC and 16.923 dB for MambaJSCC. They use independently trained latent spaces and serve only as external codec references. One training seed and a fixed 40-epoch budget do not establish convergence or a general architecture ranking. Without fine-tuning, public DeepJSCC (ImageNet/SNR 19), MambaJSCC (CLIC2021/AWGN 10), and MambaJSCC (DIV2K/Rayleigh) weights yield 13.755, 17.020, and 16.650 dB, respectively. Their protocol mismatches preclude a controlled claim about retraining benefit. Figure 1 uses solid curves for the frozen SwinJSCC latent space and dashed curves for the separately trained codecs.

C. Inference Cost and Harmful Decisions

The selector chooses Gaussian reception on 69.5% of confirmation frames. Its average cost is therefore 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. This saving does not remove harmful decisions. Against joint sampling, 3.79% of frames lose more than 0.5

TABLE II
MEAN PSNR UNDER BLOCK MISMATCH AND NO INTERFERENCE.

Profile	Direct	Gaussian	Joint	Select
Shifted blocks	20.83	22.99	21.09	23.00
Random lengths	20.64	23.42	22.74	23.98
No interference	31.37	31.37	27.73	31.37

dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret. Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318. Together, these metrics qualify the average PSNR gain. Figure 2 shows the frame-level CDF; the vertical line marks the -0.5 dB harmful-loss threshold.

D. Block Mismatch and No Interference

We test robustness on 64 additional pairs with seed 3700 and nominal SINR -4, 4, 10 dB. One profile shifts burst boundaries by half a receiver block; another draws segment lengths uniformly from 16 to 128 symbols and normalizes the binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. We report these diagnostic families separately from matched confirmation. Table II reports the robustness tests. The selector changes PSNR relative to Gaussian reception by +0.010 dB for shifted blocks and +0.563 dB for random lengths. With no interference, it matches Gaussian reception and avoids the joint receiver's loss. These are diagnostic populations; we neither refit the rule nor pool the results with matched confirmation. For a qualitative check, Fig. 3 shows all six prespecified confirmation examples: image pairs 512 and 527 under conditions 0, 8, and 15. No example was selected by reconstruction quality.

V. CONCLUSION

In this paper, we presented a blind selector that uses received energy and block-energy heterogeneity to choose between Gaussian denoising and joint signal-interference reconstruction with frozen neural weights. On independent confirmation, it improves PSNR by 0.750 dB over Gaussian denoising and 1.334 dB over joint sampling. It also uses 34.7% fewer predictor evaluations than joint sampling. The gain is specific to the evaluated population. Harmful decisions remain, and block mismatch nearly removes the advantage over Gaussian denoising. Future work will address fading, synchronization errors, and task-oriented evaluation for vehicular links.

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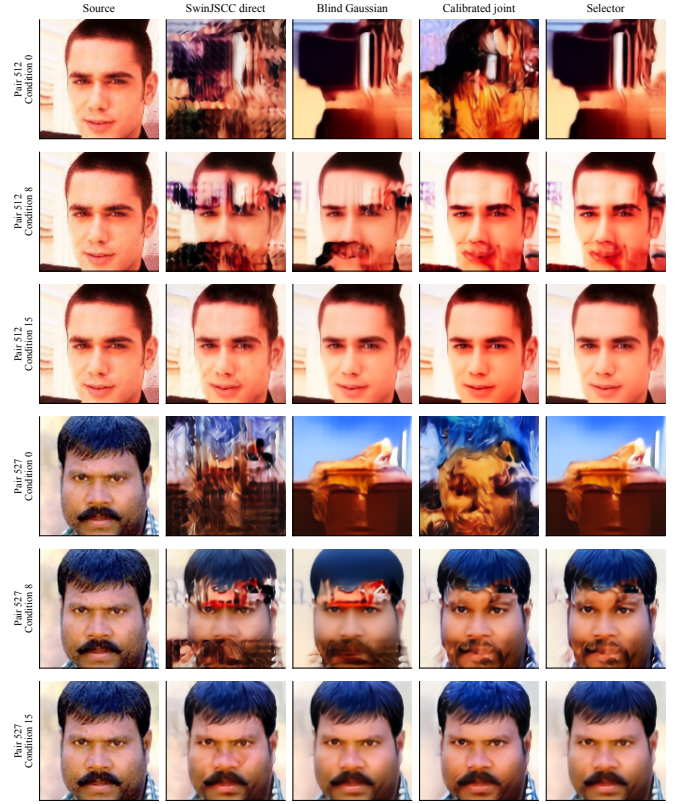


Fig. 3. Reconstructed images for six prespecified confirmation examples.

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